Recruitment CRM — Developer Specification

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Overview

This CRM runs an overseas recruitment agency end to end: a resume arrives by email or WhatsApp, is read by our OCR, lands in the candidate database, is assigned to a staff member automatically, and is then walked through 11 stages from prescreening to air ticket. Every stage hand-off becomes a task, and any task left open for more than 48 hours escalates to the admin.

The core chain the developer must build:

1. **Ingest** resumes from connected email accounts and WhatsApp, send each file to the OCR service, and save the parsed candidate.
2. **Assign** each new candidate to a staff user automatically and create the first task (initial call).
3. **Match** candidates to a client's job order from the Jobs/Positions page. Matching is what unlocks the candidate's stage modal. A candidate is mapped to one order at a time and can be unmapped and remapped to another.
4. **Process** the candidate through the stage modal, one stage at a time, with each stage unlocking only after the previous one is complete.
5. **Track** every stage as a task with a 48-hour SLA and automatic escalation.

All of this is in scope for v1.

User roles

Three roles: Admin, Manager and Staff. Staff are the recruiters who receive auto-assigned candidates and work the tasks; the Manager supervises the staff day to day; the Admin owns the system and its settings.

Capability	Admin	Manager	Staff
Connect / remove email and WhatsApp sources	Yes	No	No

Capability	Admin	Manager	Staff
Create clients, designations and orders	Yes	Yes	Yes
Map / unmap candidates to orders	Yes	Yes	Yes
See candidates in the Data Bank	All	All	All
Work the candidate stage modal	Any candidate	Any candidate	Own candidates
Reassign candidates and tasks	Yes	Yes	No
Receive auto-assigned candidates	No	No	Yes
Receive 48-hour escalations and missed-update alerts	Yes	Yes	No
Reopen a completed stage	Yes	No	No
Reports	All staff	All staff	Own numbers
User management	Yes	No	No
Personal To-do List	Yes	Yes	No

Every create, edit, stage change, reassignment and file upload is written to an audit log with user and timestamp.

Navigation and dashboard

The sidebar holds ten items, in this order: Dashboard, Inbox, Clients, Jobs / Positions, Add Candidate, Data Bank, Tasks, To-do List (admin and manager only), Reports, User Management (admin only).

The **Dashboard** is the landing page after login. It shows the logged-in user's numbers; the admin sees the whole team with a staff filter.

- **Today cards:** new candidates ingested today (email vs WhatsApp), my open tasks, tasks due today, overdue / escalated tasks.
- **Pipeline by stage:** count of active candidates sitting in each of the 11 stages, clickable to a filtered list.
- **Active orders:** each open order with client, designation, openings, candidates matched, selected, deployed.

- **My tasks:** the user's open tasks sorted by due time, overdue at the top in red.
- **Ingestion health (admin):** last sync time per connected email / WhatsApp source, and files that failed OCR.

Module specs

Inbox

One unified inbox for multiple email accounts.

- Connect accounts by Gmail OAuth, Microsoft 365 OAuth, or plain IMAP/SMTP. No limit on number of accounts.
- Unified list of all mail, with an account filter, search, unread state and attachment icon.
- Open an email, preview attachments, reply from the same account.
- Per account toggle: **Auto-ingest resumes** (on/off). When on, attachments go to the ingestion pipeline automatically.
- Manual button on any email: **Create candidate from attachment**, for mail that auto-ingest skipped.
- Each email shows a badge if it already produced a candidate, linking to that candidate.

Clients

List and form for employer clients.

- Fields: company name, country, city, industry, contact person, contact email, phone/WhatsApp, address, agreement upload, notes, status (active/inactive).
- Client detail page shows its orders and the candidates in each.

Jobs / Positions and Orders

Two levels: a **designation** is the job title (e.g. Welder, Electrician); an **order** is a specific opening for a specific client under that designation.

- Designation fields: title, category, description, required skills, minimum experience.
- Order fields (created inside a designation): client, number of openings, work country/city, salary and currency, benefits, contract period, interview date, order deadline, status (open / on hold / filled / closed).

- **Matching:** inside an order, the user searches the Data Bank (filters: designation, experience, skills, location, passport available, assigned staff) and selects candidates. Each match creates an *application* (candidate × order).
- Only candidates with an application get the stage modal. A candidate can be in only one active order at a time. To move them, the user clicks Unmap (the current application closes with reason Unmapped and its open tasks are cancelled), then maps them to the new order, which starts a fresh application at stage 1. A candidate with no match has a profile page but no stage modal.
- Order page shows matched candidates as a table with their current stage, owner and last update.

Add Candidate (entry form)

Manual entry for walk-ins, referrals and anything the automation missed. Saves to the same candidate table and triggers auto-assignment exactly like an ingested candidate.

- Basic details: full name, phone, WhatsApp number, email, date of birth, gender, nationality, current city, languages.
- Passport: number, issue date, expiry date, place of issue, **passport copy upload**.
- Documents: **resume upload** (OCR pre-fills the form when a resume is uploaded first), photo, education certificates, experience certificates.
- Work profile: preferred designation(s), total experience, Gulf/overseas experience, skills, expected salary.
- Source: email / WhatsApp / walk-in / referral / other.
- Duplicate check on save: phone, email or passport number already exists → show the existing record and ask to merge or cancel.

Data Bank

The searchable list of every candidate in the system.

- Table columns: name, phone, designation, experience, passport (yes/no), source, assigned staff, mapped order, current stage, status, date added. Every staff user can see every candidate here. A quick filter shows the Not interested pool so those candidates can be offered other relevant orders later.
- Filters on every column, full-text search on resume content, export to Excel (admin only).

- Row click opens the candidate profile; if the candidate has applications, each opens its stage modal.

Tasks

The user's task list, generated automatically (see Task engine).

- Tabs: Open, Due today, Overdue, Completed. Admin also gets an **Escalated** tab and a staff filter.
- Each task shows candidate, order, stage, assigned to, created at, due at, time remaining.
- Clicking a task opens the candidate's stage modal at that stage. Tasks are closed by completing the stage, not by a manual tick. Each open task needs a daily remark until it is completed (see Daily updates).

To-do List

A personal to-do list for the admin and managers only, separate from the system-generated Tasks. Staff do not see this menu item.

- Add an item with title (required), notes, due date and time, priority (high / normal / low), and an optional link to a candidate, client or order.
- Tabs: Today, Upcoming, Overdue, Done. Items are ticked off manually with a checkbox.
- Each user's list is private to them.
- On the due date and time, the user gets a reminder in the CRM notification bell; overdue items show in red.
- Open items also appear as a small "My to-dos" card on the admin and manager Dashboard.

Reports

All reports filter by date range, client, order, designation and staff, and export to Excel/PDF.

- Pipeline funnel: candidates per stage and drop-off between stages.
- Staff performance: candidates assigned, tasks completed, average hours per stage, escalations.
- Source report: candidates by source (each email account, WhatsApp, manual).
- Order status: per order, openings vs matched vs selected vs deployed.

- Rejections: count and reason by stage (prescreen fail, interview reject, medical unfit).
- Advance collections: amount collected per candidate and per period.

User Management

Admin only.

- Add, edit, deactivate users. Fields: name, email, phone, role, active.
- Per user toggle: **Receives auto-assigned candidates** (on/off), plus an *on leave* flag that pauses assignment.
- Reset password, force logout, view a user's audit log.
- On deactivation, prompt the admin to reassign that user's candidates and open tasks.

Resume ingestion automation

Every resume that reaches a connected email account or our WhatsApp Business number must become a saved, assigned candidate with no manual step. The pipeline runs as a background worker with a queue so a slow OCR call never blocks the app.

1. **Receive.**
   - Email: poll each account with auto-ingest on (IMAP IDLE or Gmail/Graph push where available, otherwise poll every 2 minutes). Pick up attachments of type PDF, DOC, DOCX, JPG, PNG.
   - WhatsApp: WhatsApp Business API webhook. Pick up document and image messages. Save the sender's number as the candidate's WhatsApp number.
2. **Store the original file** in object storage and log the source (account or number, message id, received time). Skip a file whose hash was already processed.
3. **Send to OCR.** Post the file to our OCR service (Veriis) and receive structured fields: name, phone, email, date of birth, nationality, passport number, current location, designation, total experience, skills, education. If the file is a passport rather than a resume, store it as the passport copy.
4. **Duplicate check** on phone, email and passport number.
   - Match found: attach the new file to the existing candidate as the latest resume, update empty fields only, and do not reassign.
   - No match: create a new candidate.
5. **Save** the candidate to the database with source and status *New*. The candidate database and the CRM database are the same; there is no separate system to sync.

6. **Quality flag.** If OCR misses name or phone, or returns low confidence, mark the candidate *Needs review* so the assigned staff checks the fields during the first call.
7. **Assign and create the first task** (next section).
8. **Failures.** OCR errors retry 3 times with backoff, then land in a *Failed ingestion* list visible to admin on the dashboard, with a manual retry button.

Optional: send the candidate an automatic acknowledgement on the channel they used (email reply or WhatsApp template message).

Automatic assignment

Every new candidate, ingested or entered manually, is assigned to one staff user the moment it is saved. The default rule is round-robin across eligible staff.

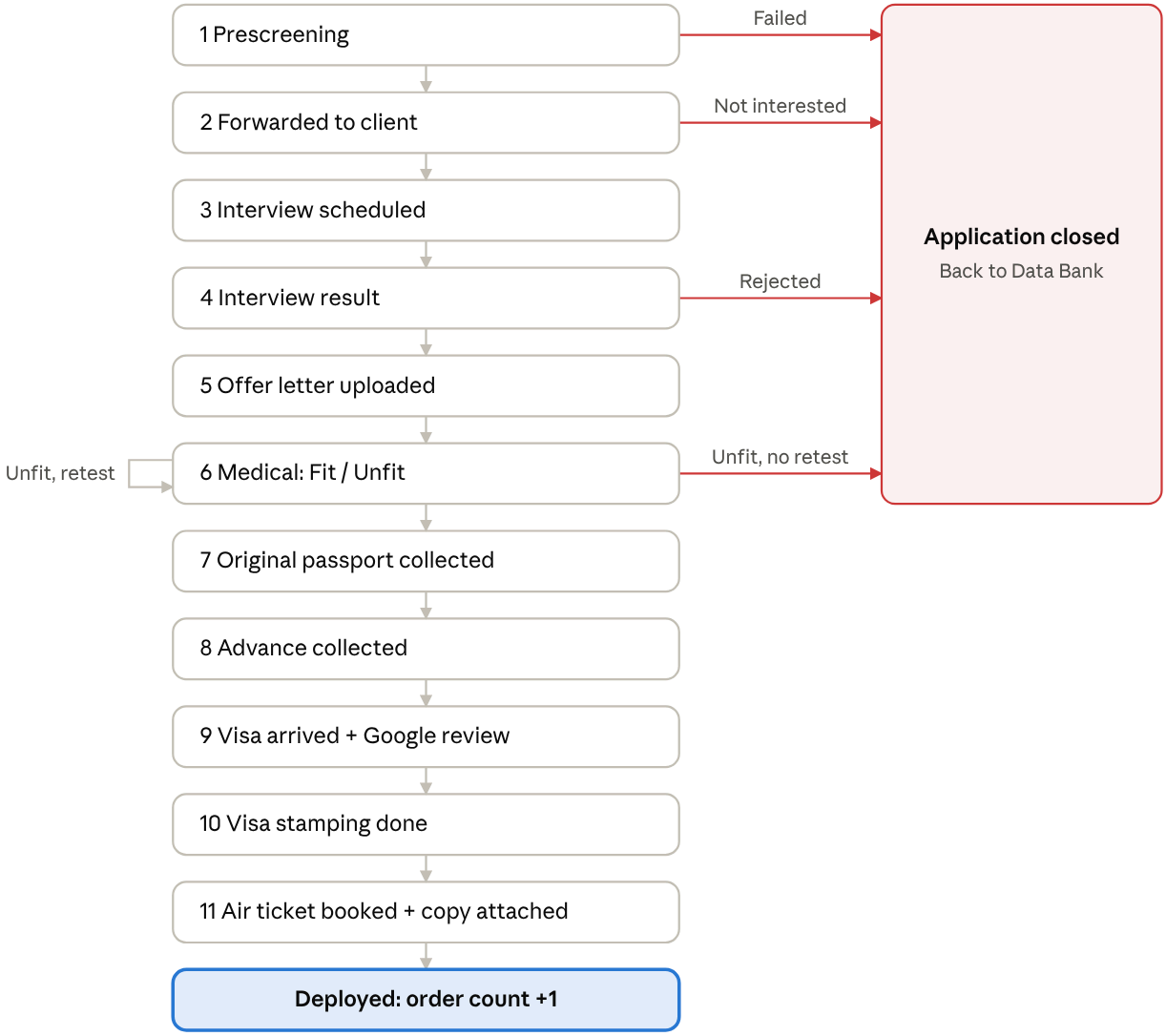
- **Eligible staff:** role Staff, active, *Receives auto-assigned candidates* on, not on leave.
- **Round-robin:** keep a pointer to the last user assigned; give the next candidate to the next eligible user in the list. Use a database lock so two candidates arriving together don't go to the same person by race. Every eligible staff member gets one candidate in turn before anyone gets a second, like a circle.
- **No load balancing:** the rotation stays strict regardless of how many open tasks a user has.
- **Duplicates** keep their existing owner.
- **Reassignment:** admin or a manager can reassign one or many candidates; all their open tasks move to the new owner and the change is logged.
- **On assignment** the system creates the first task for the owner: *Initial call — call the candidate and confirm if they are interested*. The staff records the outcome on the candidate profile: Called (yes/no), Interested (yes / no / not reachable), remarks. Not interested closes the task and marks the candidate *Not interested*; the candidate stays in the Data Bank.
- The owner and the admin are notified inside the CRM (notification bell) for each new assignment.

Candidate stage modal

The stage modal is one scrolling page with 11 stage sections, and each section appears only after the previous one is completed. It exists per application (one candidate matched to one order), so it opens from the order page, the Data Bank or a task — never for an

unmatched candidate.

Each stage unlocks the next; four outcomes close the application



candidate stage flow · 11 stages, 3 exits

Only stages 1, 2, 4 and 6 can close an application; at every other stage a "No" keeps the candidate on that stage with its task still open.

Behaviour rules

- Header: candidate name, photo, phone, passport number, client, order, designation, owner, current stage, overall status.

- Completed stages stay visible above, collapsed and read-only, showing who completed them and when. Only admin can reopen a completed stage, with a mandatory reason that is logged.
- The current stage is open and editable. Future stages are not shown.
- Clicking **Complete stage** validates the required fields, saves the stage, closes the stage's task, opens the next section and creates the next task.
- A negative outcome at stages 1, 2, 4 or 6 closes the application (status *Closed* with reason). The candidate is unmapped, returns to the Data Bank and can be remapped to another relevant order.
- Every file upload keeps version history; nothing is overwritten.

Stages

#	Stage	Staff records	To complete	Negative outcome
1	Prescreening	Result: Passed / Failed; remarks	Result selected	Failed → application closed
2	Forwarded to client	Forwarded: Yes / No; date forwarded. If No, reason: Candidate not interested / Resume not received	Yes	No (not interested or resume not sent) → candidate marked Not interested, application closed and unmapped, candidate moved to the Not interested pool for other relevant orders
3	Interview scheduling	Interview scheduled: Yes / No; date and time; mode (in person / online); venue or link	Yes + date	No → stays on this stage
4	Interview result	Selected / Rejected; interview remarks (required)	Result + remarks	Rejected → application closed
5	Offer letter	Offer letter obtained: Yes / No; offer letter upload	Yes + file uploaded	No → stays on this stage

#	Stage	Staff records	To complete	Negative outcome
6	Medical	Applied: Yes / No + date; result Fit / Unfit. If Unfit: reapplying Yes / No; if Yes, new date and new result. Re-medical can repeat.	Latest result Fit	Unfit and not reapplying → application closed
7	Passport	Passport copy (pre-filled from profile, replaceable); original passport collected: Yes / No; date collected; held by (staff name)	Collected = Yes	No → stays on this stage
8	Advance payment	Advance collected: Yes / No; amount (entered per candidate, varies); date; payment mode; receipt number; receipt upload	Yes + amount + receipt	No → stays on this stage
9	Visa arrival + Google review	Visa arrived: Yes / No; date; visa copy upload. Google review obtained (checkbox)	Visa arrived = Yes	Visa not arrived → stays on this stage
10	Visa stamping	Stamping done: Yes / No; date; stamped visa copy upload	Yes + file	No → stays on this stage
11	Air ticket	Booked by: Employer / Agency; travel date; airline and flight no.; PNR; ticket copy upload (required)	Ticket uploaded	—

Completing stage 11 sets the application to **Deployed** and the order's deployed count increases by one. When deployed equals openings, the order moves to *Filled*.

Task engine and escalation

Tasks are created and closed only by the system: each event below creates one task for the candidate's owner with a 48-hour due time, and the task closes automatically when the matching action is saved.

Event	Task created	Closes when
Candidate assigned	Initial call: confirm interest	Call outcome recorded
Candidate matched to an order	Stage 1: prescreen the candidate	Stage 1 completed
Stage N completed (N = 1 to 10)	Stage N+1 with its own title, e.g. "Forward CV to client", "Collect original passport"	Stage N+1 completed
Medical result Unfit with reapply = Yes	Follow up re-medical on the new date	New result recorded
Stage 11 completed	None; application Deployed	—
Application closed	All its open tasks cancelled	—

Escalation

- A scheduler runs every 15 minutes and finds open tasks where now > created + 48 hours.
- Each one is marked *Escalated* and the admin and managers are notified inside the CRM (notification bell), with candidate, stage, owner and hours overdue.
- One escalation per task by default; if it stays open, a reminder repeats every 24 hours (admin setting).
- The owner also sees the task in red in their Tasks and Dashboard. The 48-hour clock never pauses, including while waiting on the client or the embassy; that waiting is recorded in the daily task remarks.
- Admin or a manager can reassign an escalated task from the notification. The 48-hour clock does not reset on reassignment unless the admin chooses to.
- Every escalation is stored and counted in the staff performance report.

Daily updates (mandatory)

Every open task must get at least one remark from its owner each day until the task is completed. The remark says what happened that day, e.g. "Called, candidate will send resume tomorrow" or "Waiting for client feedback".

- Each task has an **Add remark** box (text, required) in the Tasks list and inside the stage modal. Remarks are timestamped, cannot be edited or deleted, and show as a history under the task.
- A task with no remark for the current day shows a *Not updated today* badge in the owner's Tasks and Dashboard.
- At end of day, the system lists every open task that got no remark that day and notifies the admin and managers inside the CRM, grouped by staff.
- Completing the stage closes the task; no remark is needed on the day it is completed.
- Missed daily updates are counted per staff in the staff performance report.